



CNR

CRAFT N ROAM

AI Travel Decision Simulator & 3D Walk

Craft a transparent plan. Roam the winner in 3D.

Nonthapat T. · Jeeraporn Y. · Chaiphan N. · Jirun T.

Generative AI · Final Project Presentation

The Problem

Trip planning means balancing budget, time, and taste across dozens of options — and every existing tool skips the reasoning.



Listing sites

Google Maps, TripAdvisor, Yelp show everything and decide nothing.



AI chatbots

One suggestion, no visible tradeoffs — a black box.



Either way

No way to get a real feel for a place before you commit.

Craft, Then Roam

A transparent, traveler-adjustable decision engine, followed by a review-grounded 3D preview of the winner.



1. Craft the plan

- Traveler describes their trip in free text (69 tokens)
- LLM parses it into structured weights
- Deterministic engine scores & ranks 5 curated spots
- Sliders let travelers fine-tune live, for free



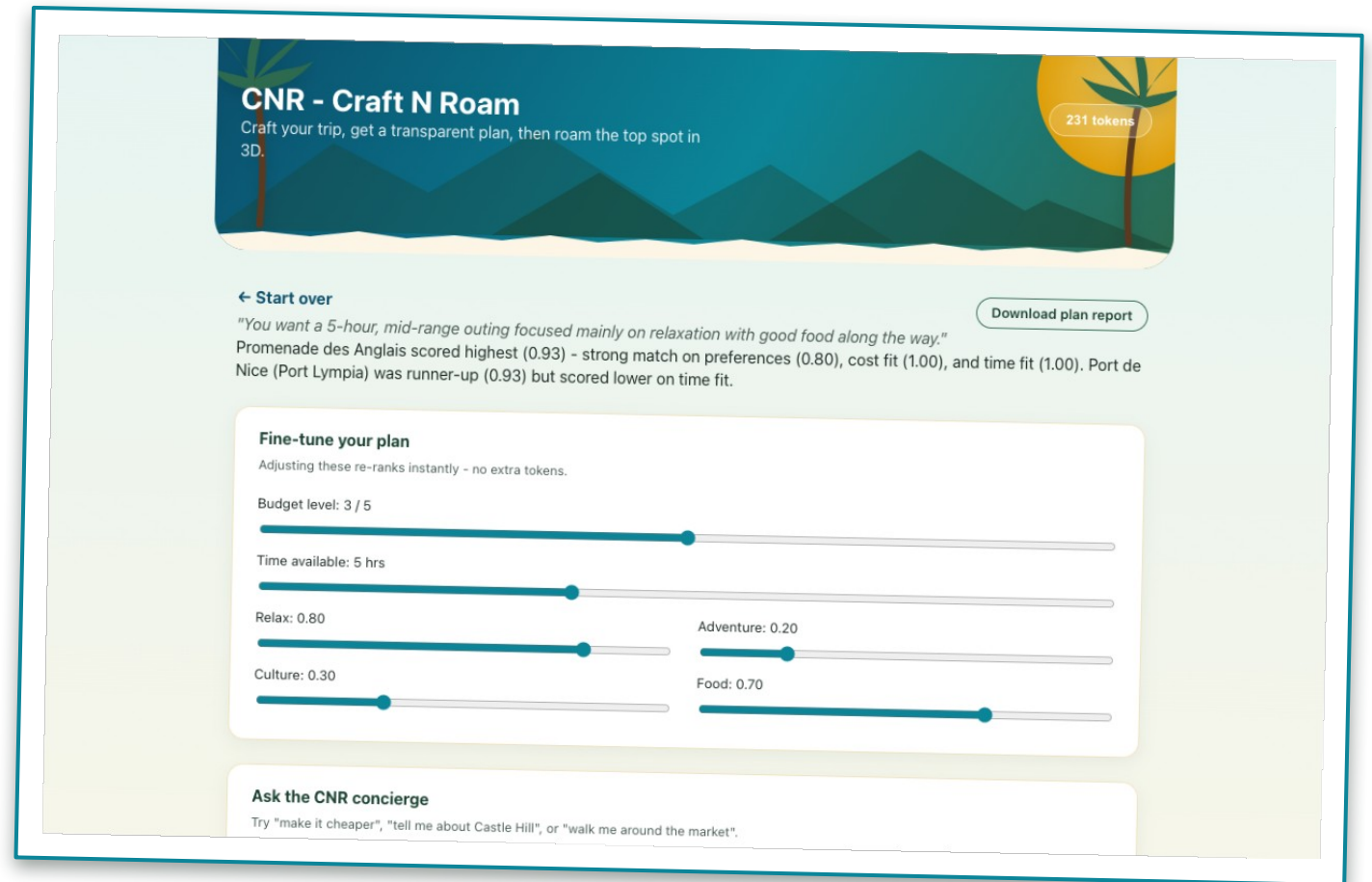
2. Roam in 3D

- Traveler unlocks all spots in 3D once (89 tokens)
- LLM reads that spot's curated reviews
- Extracts ground / sky / palette / props (fixed catalog)
- Seeded layout builds a walkable Three.js scene

Demo — Craft a Transparent Plan

"A relaxed five-hour trip, mid-range budget, good food"

- Promenade des Anglais scores 0.93 — strong match, perfect cost & time fit
- Sliders adjust budget, time, and category balance
- Re-ranks instantly — no extra tokens, no black box



Demo — Roam the Winner

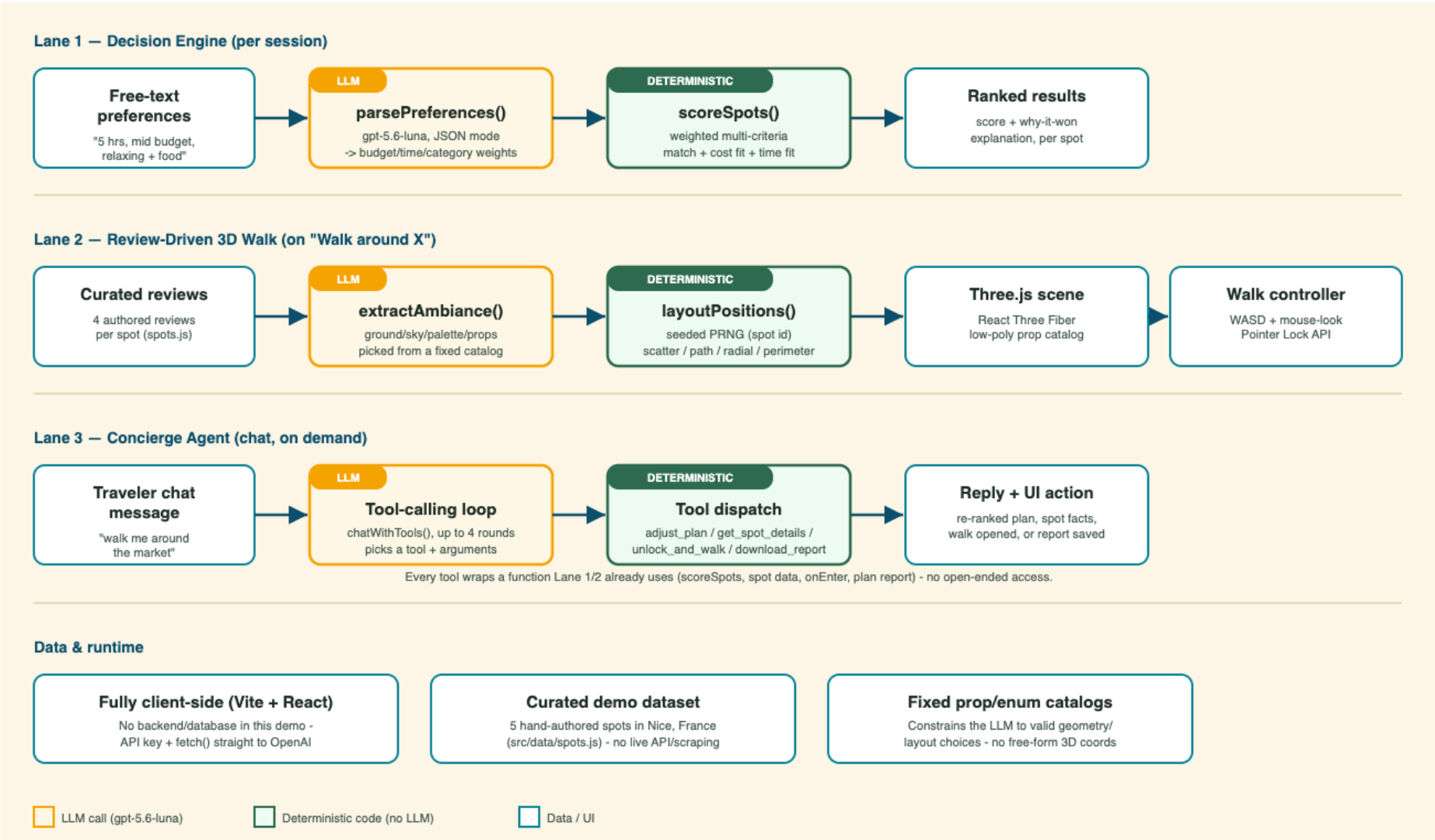
- LLM reads Promenade des Anglais' reviews
- Picks ground, sky, palette, and props from a fixed catalog
- A seeded algorithm places everything — same input, same scene
- Explorable in first person: WASD + mouse-look
- Share 3 photos + a review per spot, earn 30 tokens back



Stylized, not photorealistic — by design (see limitations).

How the AI Is Used

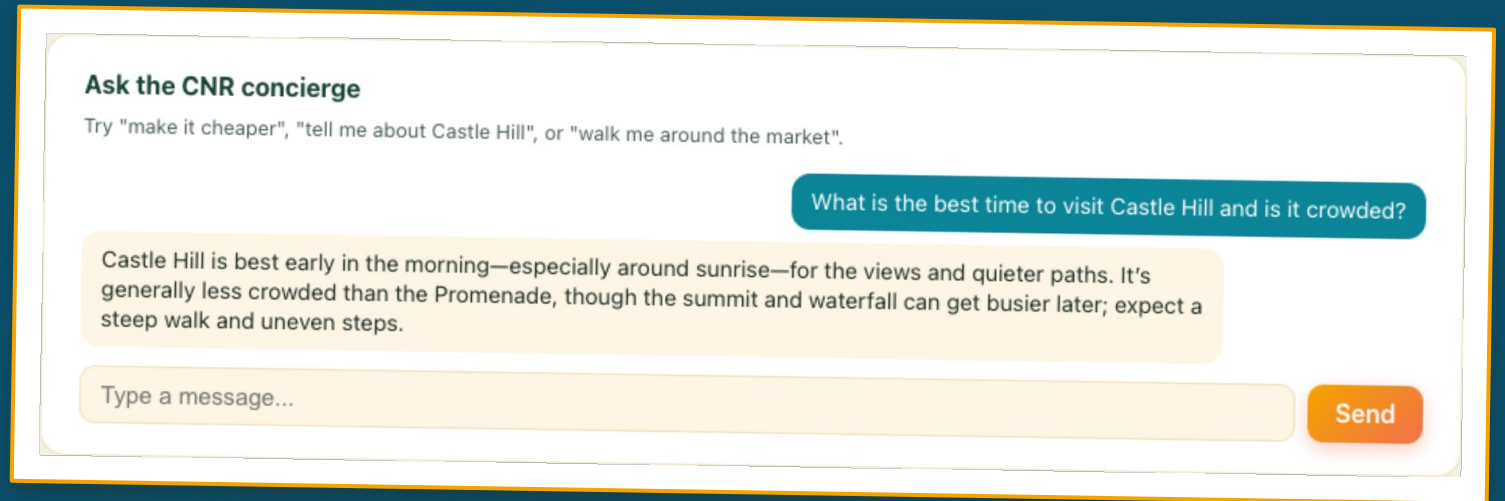
Three lanes of narrow LLM steps; every decision and every agent tool call bottoms out in ordinary deterministic code.



Demo — Ask the Concierge (Lane 3)

A real tool-calling agent - four tools, each a thin wrapper around a function the app already trusted.

- `adjust_plan` - "make it cheaper" re-ranks live
- `get_spot_details` - answers grounded in curated reviews
- `unlock_and_walk` - "walk me around the market" opens the 3D scene
- `download_report` - saves the plan report



Capped at 4 tool-call rounds per turn - no runaway loops.

One Design Principle

Lanes 1-2: the LLM translates, it never decides. Lane 3 may act - but only through tools that wrap trusted code.



LLM calls

- Free text -> structured weights
- Reviews -> structured ambiance spec
- Fixed JSON schema, fixed enum catalogs
- Validated & clamped before use



Deterministic code

- Weighted multi-criteria scoring
- Live plan-editor re-ranking
- Seeded PRNG for 3D layout
- Same inputs -> same outputs, always

A Real Integration Bug We Hit

gpt-5.6-luna rejects function-tool calling combined with its default reasoning_effort on /v1/chat/completions.

```
invalid_request_error - param: reasoning_effort
```

```
"Function tools with reasoning_effort are not supported for gpt-5.6-luna in /v1/chat/completions."
```

Fix: request raw JSON (response_format: json_object) instead of tool-calling, and pass reasoning_effort: "none" explicitly.

One Honest Limitation

The 3D scenes are stylized, low-poly interpretations — not photorealistic reconstructions of the real locations.

True 3D reconstruction needs dozens of calibrated, overlapping photos (photogrammetry / NeRF) — not review text or a handful of casual photos. We chose to be upfront about this rather than overclaim, and built a deterministic, review-grounded stylized scene instead. The token economy is the same story: presented in full, built as local state only — no real payment happens anywhere in this demo.

Also flagged: WASD movement needs a final real-browser verification pass (automated headless testing was inconclusive due to background-tab throttling).



The Business Model (Presentation Only)

Simulated in the demo as local app state — no real payment processing exists today.

Item	Price
Small token bundle	\$0.80 -> 100 tokens
Large token bundle	\$4.00 -> 600 tokens (~17% off)
Generate a plan (search)	69 tokens (~\$0.46-0.55)
Unlock 3D walking (all spots)	89 tokens (~\$0.60-0.71, one-time)
Share 3 photos + a review, per spot	+30 tokens back

Margin check

Actual LLM cost per search: a fraction of a cent. Priced at 69 tokens (~\$0.50), that is roughly two orders of magnitude of margin before hosting, support, and refunds.

What's Next



Headline idea: crowdsourced photos, real rewards

Travelers upload up to 3 of their own photos per spot -> vision model extracts richer ambiance -> traveler earns a real 30-token reward per spot. Fully legitimate (user-submitted, not scraped). Needs accounts, storage, and moderation — a real backend, deliberately out of scope for now.



Roadmap timeline (details in the report)

Thank you

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<https://github.com/GongNT/travel-3d-simulator>

